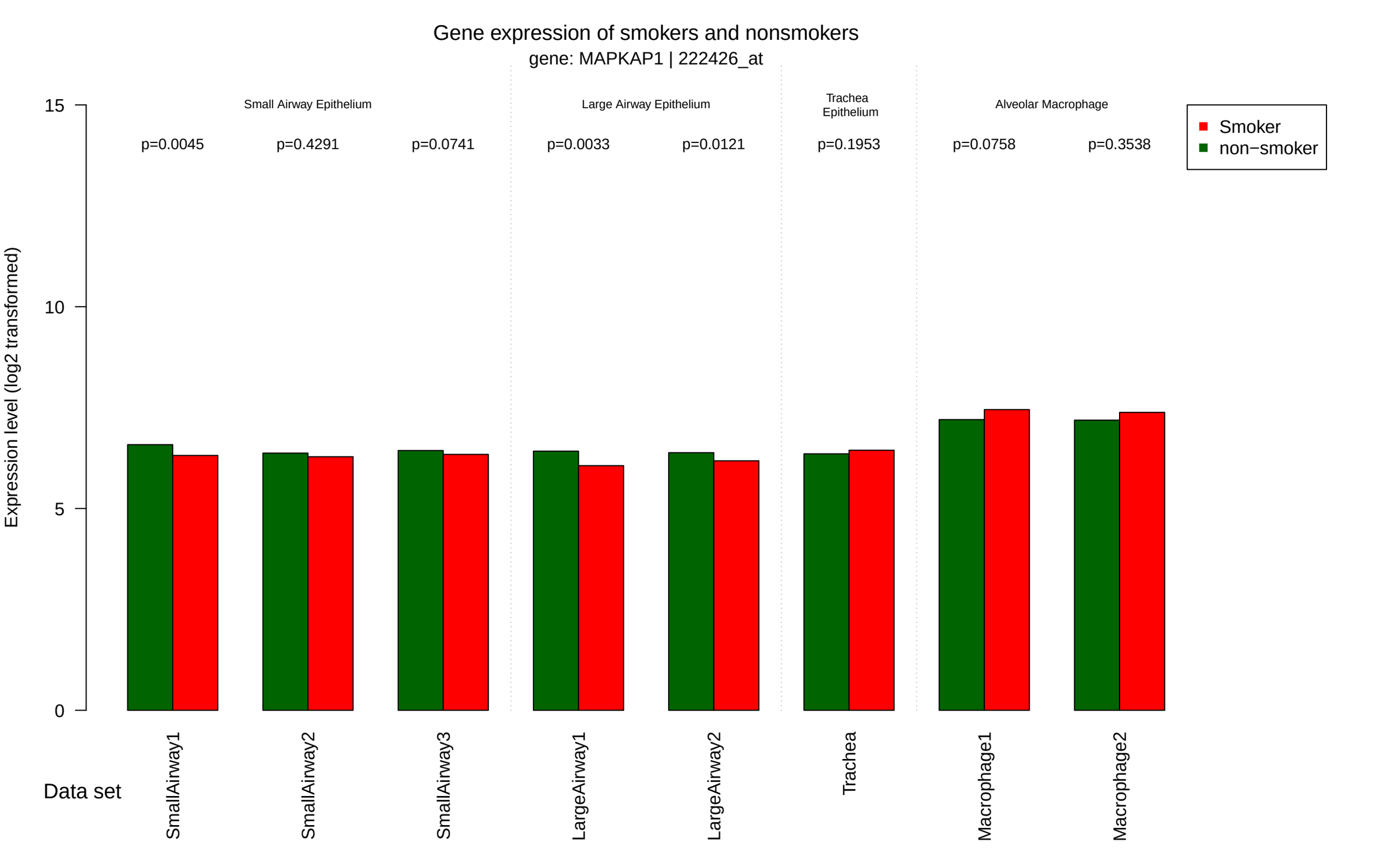
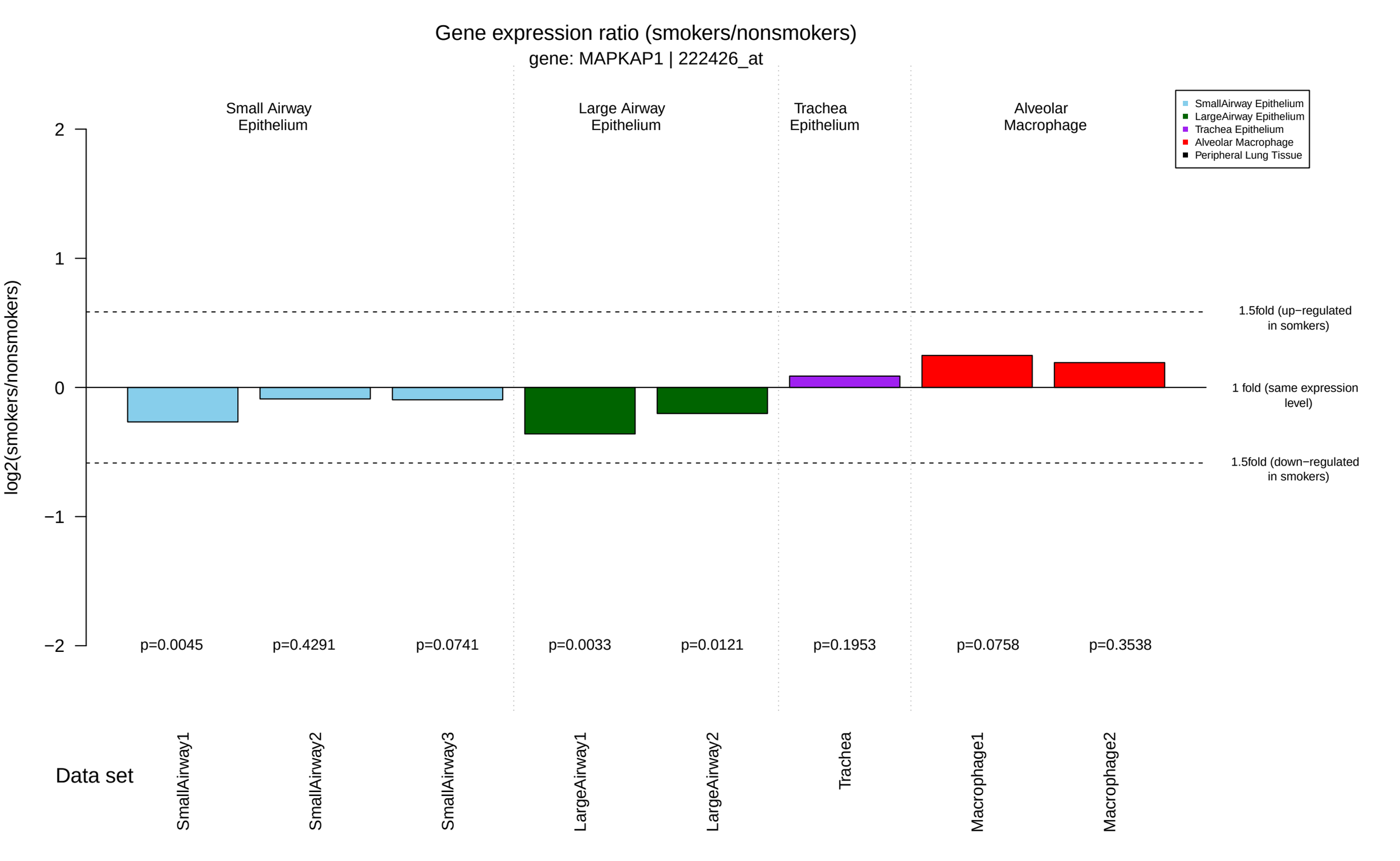
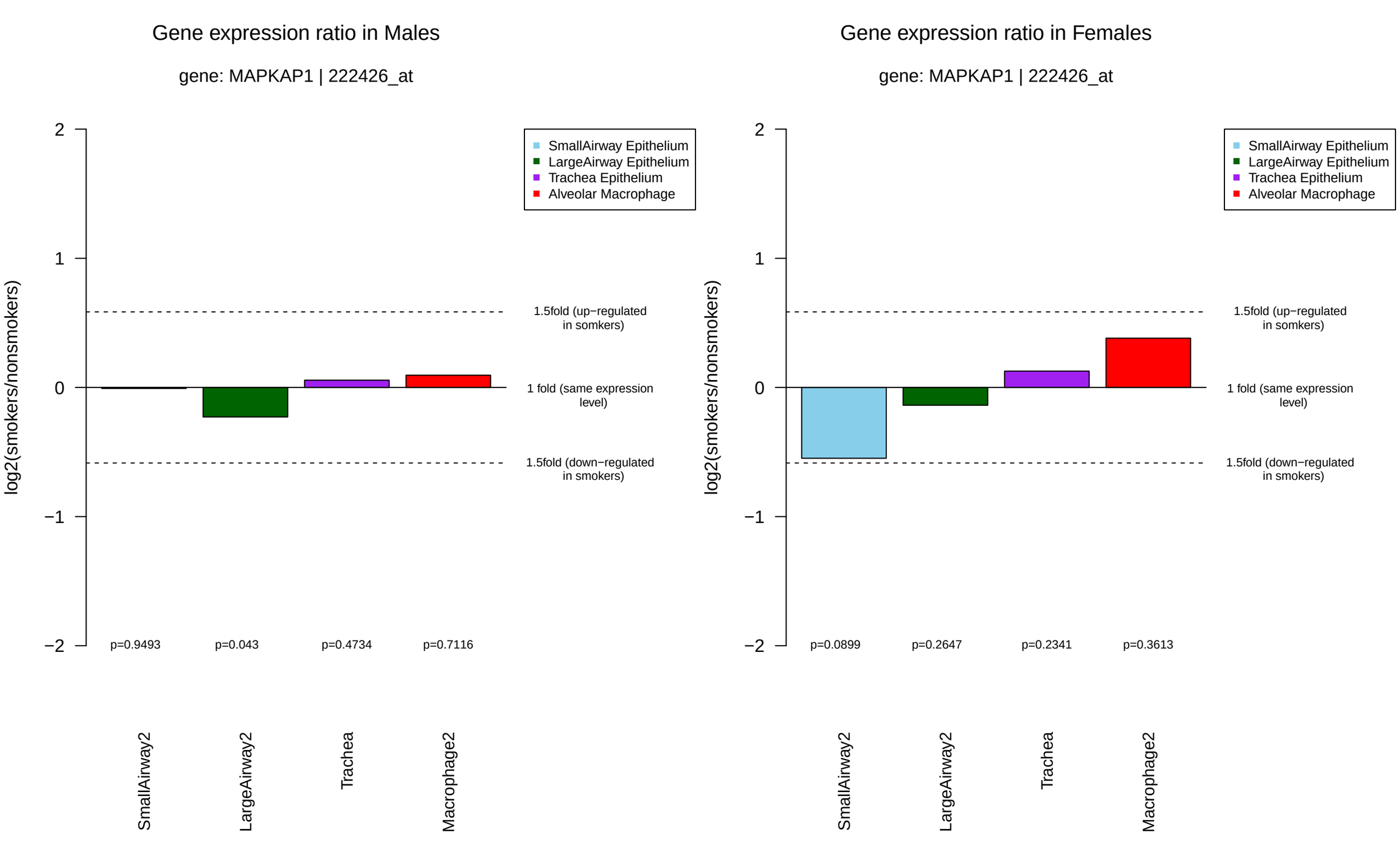
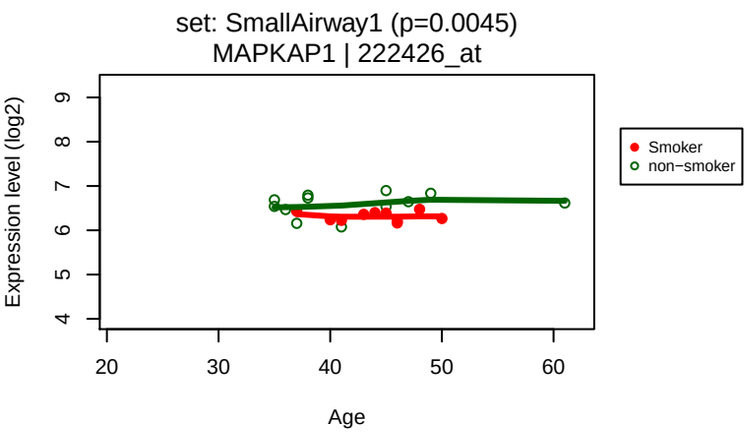
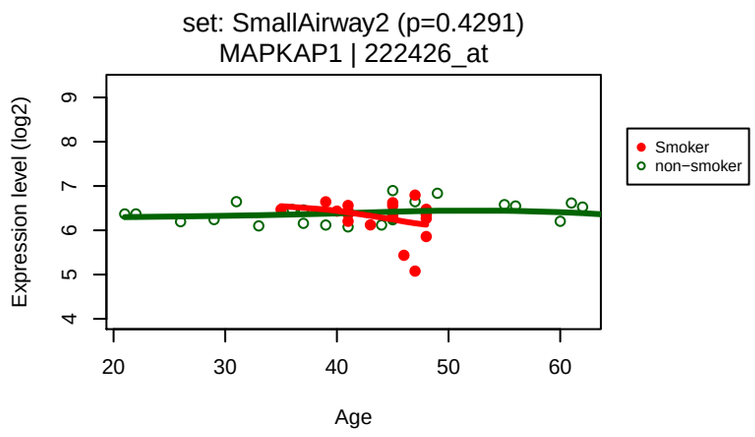
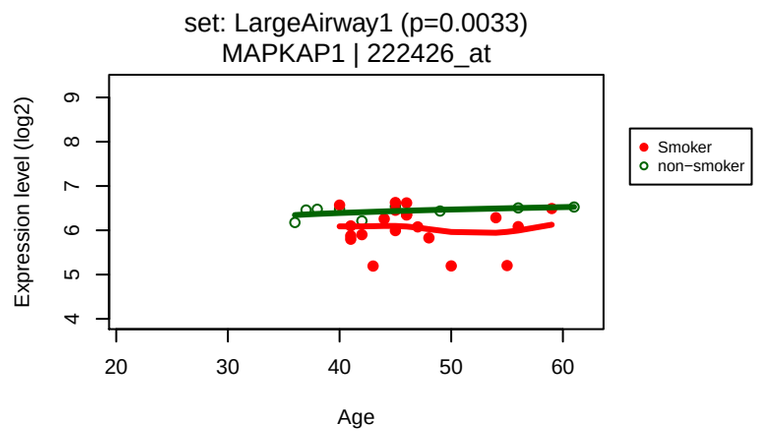

Gene expression of smokers and nonsmokers
gene: MAPKAP1 | 222426_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0045
p=0.4291
p=0.0741
p=0.0033
p=0.0121
p=0.1953
p=0.0758
p=0.3538
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

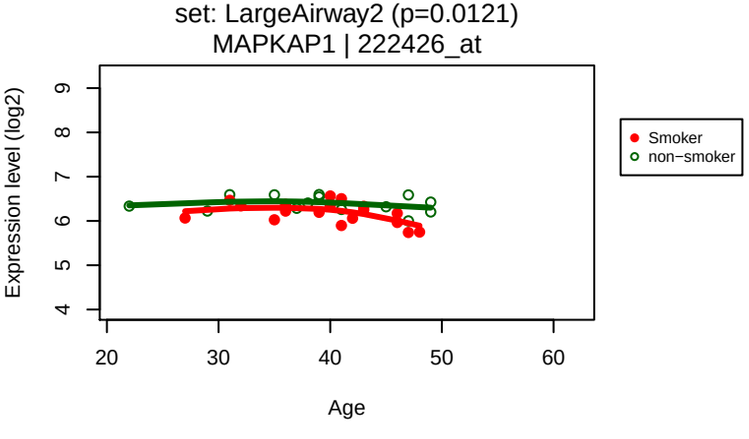

Gene expression ratio (smokers/nonsmokers)
gene: MAPKAP1 | 222426_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0045
p=0.4291
p=0.0741
p=0.0033
p=0.0121
p=0.1953
p=0.0758
p=0.3538
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

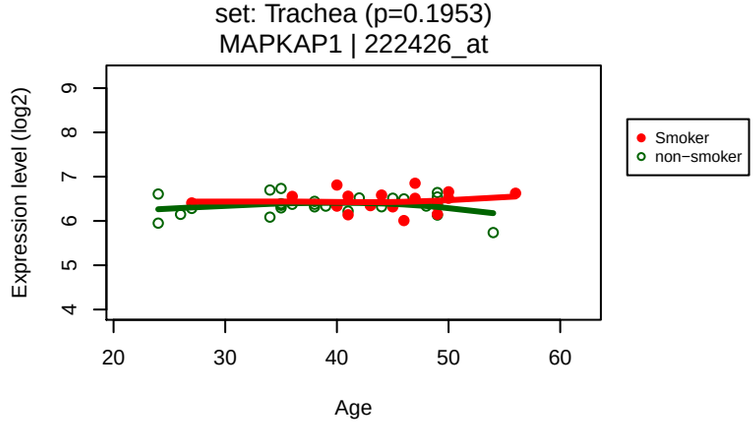

Gene expression ratio in Males
gene: MAPKAP1 | 222426_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9493
p=0.043
p=0.4734
p=0.7116
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MAPKAP1 | 222426_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0899
p=0.2647
p=0.2341
p=0.3613
SmallAirway2
LargeAirway2
Trachea
Macrophage2

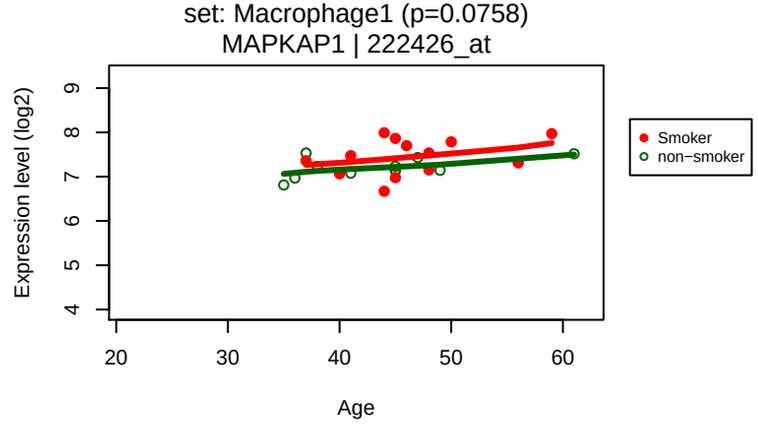

set: SmallAirway1 (p=0.0045)
MAPKAP1 | 222426_at
Expression level (log2)
Age
Smoker
non−smoker

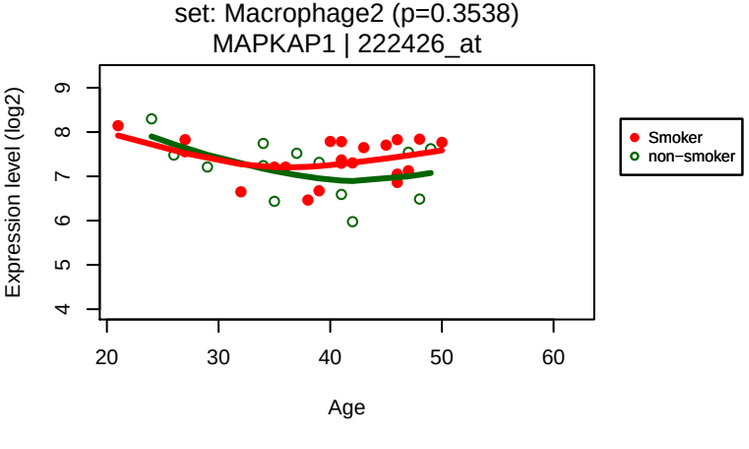

set: SmallAirway2 (p=0.4291)
MAPKAP1 | 222426_at
Expression level (log2)
Age
Smoker
non−smoker

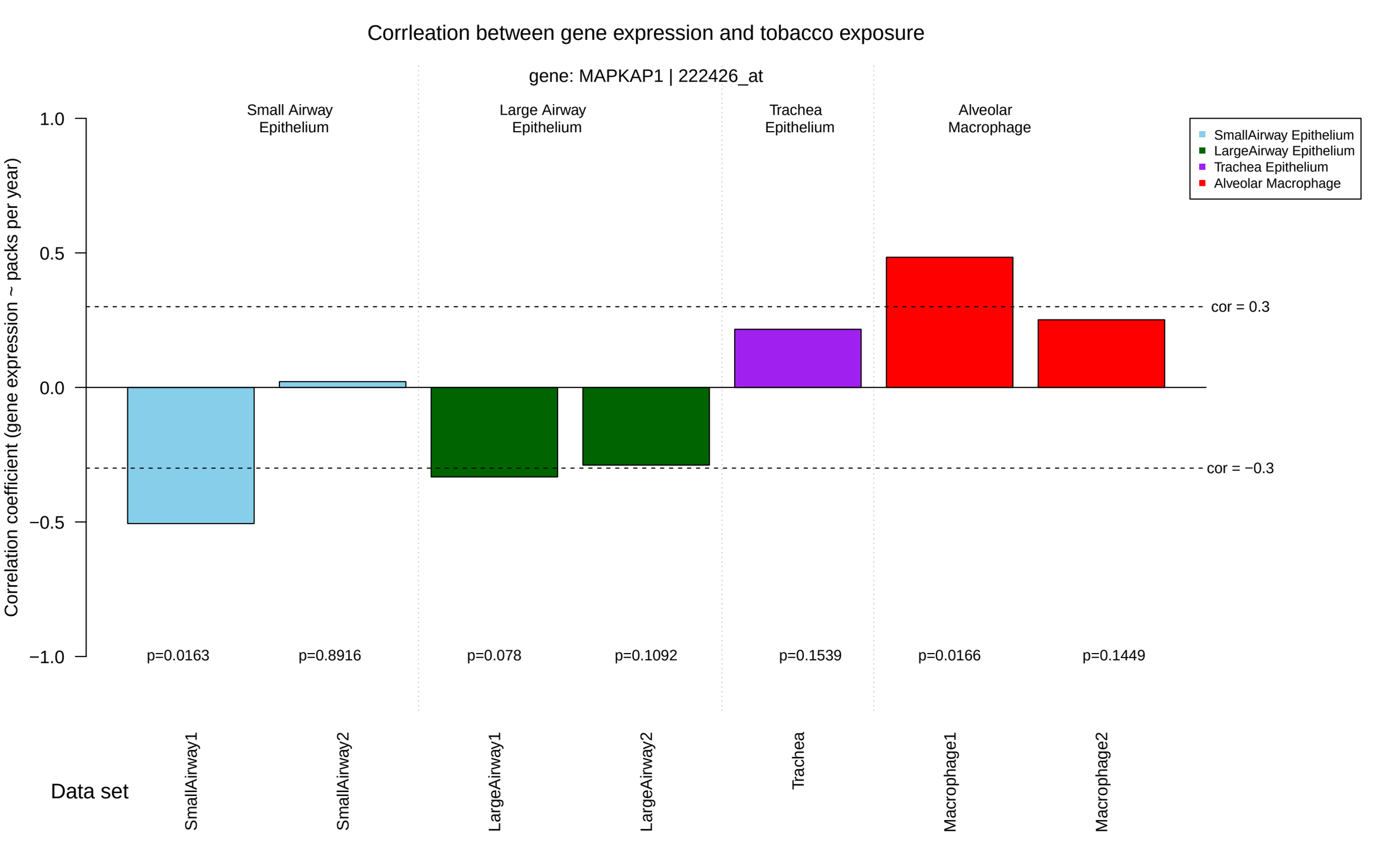

set: LargeAirway1 (p=0.0033)
MAPKAP1 | 222426_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0121)
MAPKAP1 | 222426_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1953)
MAPKAP1 | 222426_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0758)
MAPKAP1 | 222426_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3538)
MAPKAP1 | 222426_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MAPKAP1 | 222426_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0163
p=0.8916
p=0.078
p=0.1092
p=0.1539
p=0.0166
p=0.1449
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2